

GaN Device Modelling Using Machine Learning Approach for Power Electronic Converters

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Abstract— This paper introduces a deep learning-based approach for modelling various GaN-based power electronic devices, crucial for designing and simulating diverse power electronic circuits and conducting performance analysis. The deep learning model is trained using different voltage, current levels, gate resistances, and parasitic inductances. The training dataset is obtained by simulating a realistic Double Pulse Test (DPT) circuit, considering parasitic elements. A many to many Long Short-Term Memory based deep learning model is designed to effectively capture complex switching behavior of GaN devices. This model enables efficient analysis of switching voltage and current, along with reliability assessment of GaN-based power converters. The accuracy of the proposed model is verified by comparing the LTspice simulation results.

Keywords—Deep learning-based approach, Gallium Nitride (GaN), Double Pulse Test (DPT), Parasitic elements, Long Short Term Memory (LSTM).

I. INTRODUCTION

The excessive use of Internal Combustion Engine (ICE) vehicles rapidly increases pollution, leading to a rise in the demand for electric vehicles (EVs) nowadays [1,2]. One of the most important components of EVs is power converters. Currently, gallium nitride (GaN) is preferred for converters as it provides a wider energy gap, higher saturated electron velocity, and greater electron mobility [3]. Therefore, analysing the performance of GaN devices is necessary.

Analytical modelling is used to assess critical factors such as false triggering, switching loss, and lifetime analysis [4-6]. State space expressions are necessary to establish the analytical model, which is challenging and time-consuming. Developing accurate models for these assessments requires in-depth knowledge of the physical properties and behavior of GaN devices under different operating conditions.

To address these challenges and improve the efficiency of modelling, a Long Short-Term Memory (LSTM) based deep learning model is proposed. LSTM networks are a type of recurrent neural network (RNN) that are particularly well-suited for sequence prediction problems due to their ability to learn long-term dependencies. By leveraging the capabilities of LSTM networks, the proposed model aims to provide more accurate predictions of GaN device performance while significantly reducing the time and effort required for the analysis.

To estimate the damage per accumulation for each switching cycle of GaN devices, it is necessary to find the values of peak current (I_{DP}) and blocking voltage (V_{DD}). These values can be obtained from the switching locus of drain-to-source current (i_{ds}) versus switching voltage (v_{ds}) plot. Once these values are determined, they are substituted into the

TABLE I: COMPARISON OF EXISTING MODEL AND PROPOSED MODEL

	Supply Voltage (V_{dc})	Source Current (I_{dc})	Gate Resistance (R_g)	Stray Inductance (L_{stray})	Track Resistance (R_{track})
[8]	✓	✓	✗	✗	✗
[9]	✓	✓	✗	✗	✗
Proposed Model	✓	✓	✓	✓	✓

expression (1) [7], which will provide the switching lifetime of the GaN device under different operating conditions. This process is crucial for understanding and predicting the longevity and reliability of GaN devices in various applications.

$$\Delta\tau(V_{DD}, I_{DP}) = \frac{1}{A} e^{(\beta_V V_{DD} + \beta_C I_{DP})} \quad (1)$$

Where, $\Delta\tau$ denotes the damage accumulation per switching cycle, β_V is assigned a value of 0.039, and β_C is set to 0.47. Even for LTspice simulations, such as charging a battery, calculating the damage per accumulation of each high-frequency switching cycle requires a significant amount of time. This process involves complex computations and detailed analysis, which can be quite time-consuming. However, by using the proposed approach, this time can be significantly reduced. The new method streamlines the calculations and enhances efficiency, making it easier to predict the longevity and reliability of GaN devices in various applications.

Various machine learning algorithms have been used to estimate switching transitions [8,9], but they often overlook essential factors like parasitic elements and variations in gate resistance, making them inadequate as replacements for analytical models. In contrast, our proposed deep learning model integrates PCB parasitic inductance, gate resistance, supply voltage variation (V_{dc}), and load current (I_{dc}) to accurately replicate switching transitions. Moreover, this model is employed to assess the switching lifetime of GaN devices under diverse operating conditions.

Table 1 presents a comparative analysis between the existing models and our proposed deep learning approach, demonstrating its superior performance in accounting for these crucial real-world factors. By considering these elements, our model enhances the precision and reliability of predicting switching behavior, thereby addressing limitations in current analytical and machine learning-based methodologies. This advancement is critical for optimizing the performance and longevity of GaN devices in practical applications.

The main contributions of this paper are summarized as follows:

1. Employing LSTM to enhance GaN switch transition modelling for converters, demonstrating its capability in accurately predicting switching waveforms and capturing device behavior.
2. Exploring the potential of LSTM models to optimize power electronic systems, addressing the demand for high-efficiency, high-performance devices across various applications.
3. Emphasizing the consideration of a limited number of parasitic elements in the modeling process, enhancing the accuracy and applicability of the proposed LSTM-based approach.

The structure of this paper is as follows: the advantages of LSTM over other neural networks and the basics and structure of LSTM are presented in Section II. The simulation approach for this paper and the corresponding simulation results are shown in Section III. The comparison of experimental results versus simulation results is provided in Section IV.

II. MODEL AND ALGORITHM

Switch transitions and oscillations in circuits are influenced by parameters like gate resistance, loop inductance, source inductance, and drain inductance, leading to varying turn ON and turn OFF times for the device. Hence, it's crucial to develop a model capable of handling such time-series data effectively. This model needs to accurately capture the dynamics of the turn-on transition, as it plays a significant role in determining cumulative damage accumulation in the device.

A. Description of the model for switching transition estimation:

A neural network is a computational model inspired by the human brain's structure and function. It consists of interconnected nodes (neurons) organized in layers. Each neuron receives inputs, processes them through an activation function, and passes an output to the next layer. Neural networks are trained using algorithms that adjust the weights of connections between neurons to minimize prediction errors during tasks such as classification or regression [10].

For time-series data, traditional neural networks struggle to effectively capture temporal dependencies because they treat each input independently. This limitation prompted the shift towards Recurrent Neural Networks (RNNs). RNNs are specialized neural networks designed to process sequences of data where each input is not only influenced by its current state but also by preceding inputs. This makes RNNs suitable for tasks involving time-series data, such as predicting future values based on past observations.

However, RNNs face challenges with long-term dependencies due to the vanishing gradient problem, where gradients diminish as they propagate back through time [11]. To address this issue, Long Short-Term Memory (LSTM) networks were developed. LSTMs incorporate memory cells and gating mechanisms that allow them to selectively retain or forget information over extended sequences, making them more effective than traditional RNNs for modelling and predicting complex temporal relationships in data.

Therefore, an LSTM model is chosen for generating the switching transition, as it efficiently manages time-series data. This section demonstrates the data-driven deep learning

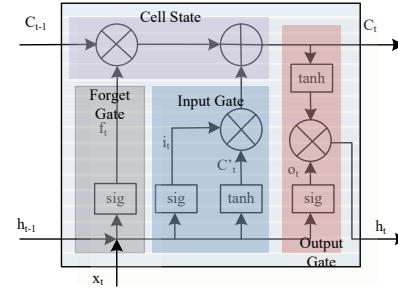


Fig. 1. Structure of LSTM block

model used to generate the switching transition of the GaN device. LSTM is a special type of recurrent neural network (RNN) with four processing modules that effectively capture the long-term dependencies of training data [12]. A many-to-many LSTM model is selected for generating the switching transition. In this model, several LSTM blocks are connected one after another, and each block is dependent on the previous blocks. The number of blocks depends on the number of samples in a training dataset. The structure of one LSTM block is shown in Fig. 1. In this figure only one LSTM block is shown.

The structure has four gates that control the flow of information i.e. training data, and their expressions can be described as follows:

Forget gate (f_t) expression:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

Input gate (i_t) expression:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

$$C'_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (4)$$

Cell state (C_t) stores and transfers updated information:

$$C_t = f_t \circ C_{t-1} + i_t \circ C'_t \quad (5)$$

Output gate (o_t) computes the output:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t \circ \tanh(C_t) \quad (7)$$

where, W_f, W_i, W_c, W_o : weight matrices, b_f, b_i, b_c, b_o are bias vectors, $\circ$ represents element wise product, h_t is state output vector and C'_t is candidate cell state. Here, x_t is called the input feature matrix. These input features (x_t) of the LSTM model are input supply voltage (V_s), input current (I_s), gate resistance (R_g), drain inductances (L_d), track resistance (R_{track}) and switching time. Output class the LSTM model is drain to source voltage (v_{ds}), drain to source current (i_{ds}), since the objective is to generate the switching transition.

B. Training data generation:

The LSTM model requires a substantial amount of training data to accurately set its parameters. The LTSpice simulation software is widely used in power electronics because its results are very close to experimental outcomes due to its use of behavioral models. Therefore, LTSpice is utilized to generate the training data.

A Double Pulse Test (DPT) circuit is simulated multiple times for different values of parameters, as presented in Table I. The switching transitions are captured for these various

parameter values and used as training data. The turn-on transition [7] of the switch for training is captured to calculate the damage accumulation per switching cycle.

The DPT (Fig.2) circuit is used in this model because the hard-switching transitions are consistent across different circuits, as reported in [13]. The simulation automation is performed using the method reported in [9].

III. SIMULATION RESULTS

To ensure realism in the simulation, the manufacturer recommended LTspice switch model GS66508B, and gate driver ADUM4121ARZ are utilized (Fig. 2). In this simulation, the turn ON transients are captured for different values of PCB parasitic inductances, input DC bus voltages and currents. The parasitic elements are taken into account

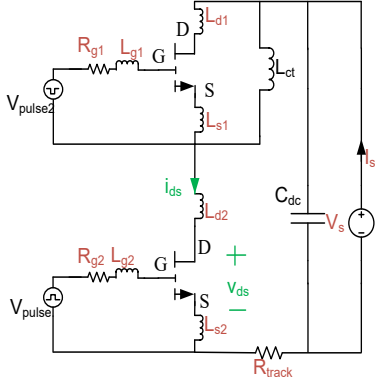


Fig. 2: Schematic of DPT circuit with main parasitic elements

since they affect the switching transition and hence the switching reliability as expressed in (1).

The Fig. 2 shows the main parasitic elements along with the necessary parameters used for training the LSTM model. The v_{ds} and i_{ds} are captured from the lower side switch of the DPT circuit, used as training and testing data for the LSTM model. The supply voltage (V_s), input current (I_s), gate resistance (R_g), drain inductances (L_d), track resistance (R_{track}) are varied in a specified range for generating the training data. The range of values for different parameters used to train the LSTM model is comprehensively represented in Table II.

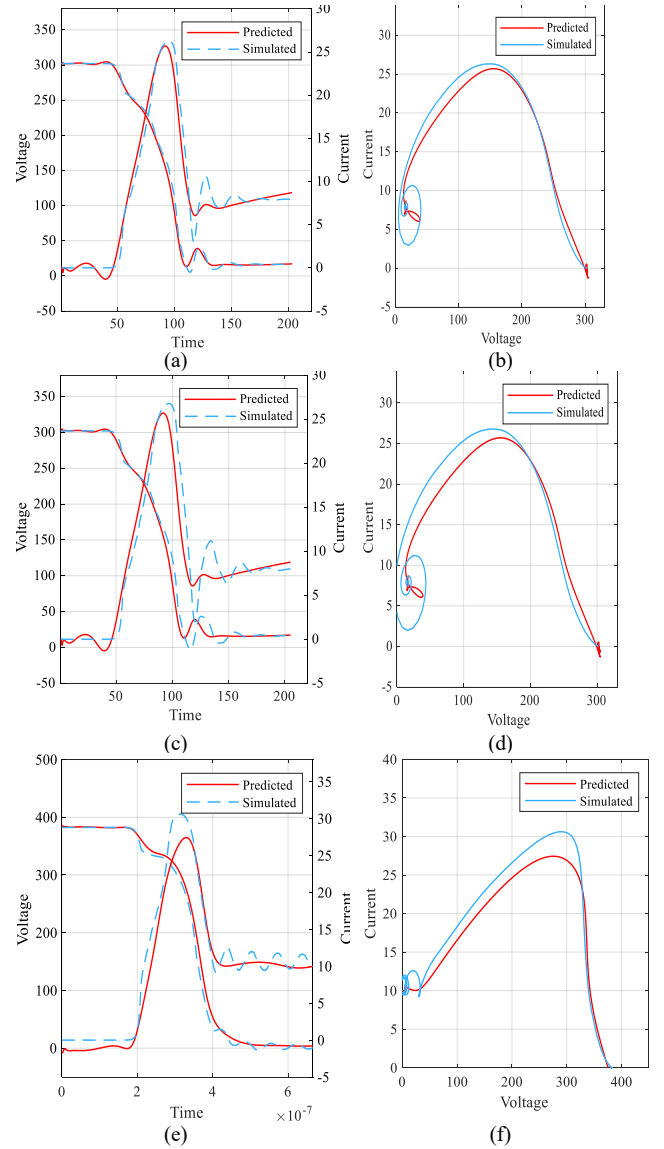
TABLE II: TRAINING RANGES OF DIFFERENT PARAMETERS

Parameters	Ranges
V_s (V)	50-150
I_s (A)	6-8
R_{g1} (Ω)	10-20
R_{g2} (Ω)	10-20
L_{d1} (nH)	0.5-2.5
L_{d2} (nH)	0.5-2.5
L_{s1} (nH)	0.5-2.5
L_{s2} (nH)	0.5-2.5
L_{g1} (nH)	1-3
L_{g2} (nH)	1-3

The generated training data is imported to the Matlab for training the LSTM Model. Results during the turn-on transition of the GaN switch are observed from the proposed trained LSTM and verified for accuracy using four different datasets as outlined in Table III.

TABLE III: DIFFERET SETS RESULTS

Parameters	Set 1	Set 2	Set 3	Set 4
V_s (V)	300	300	380	380
I_s (A)	7	7	9	9 A
R_{g1} (Ω)	20	20	24	28
R_{g2} (Ω)	20	20	24	28
L_{d1} (nH)	2	1.75	1.75	1.75
L_{d2} (nH)	2	1.75	1.75	1.75
L_{s1} (nH)	2	1.75	1.75	1.75
L_{s2} (nH)	2	1.75	1.75	1.75
L_{g1} (nH)	1	1	1	1
L_{g2} (nH)	1	1	1	1
R_{track} (Ω)	2	2	0.2	0.2



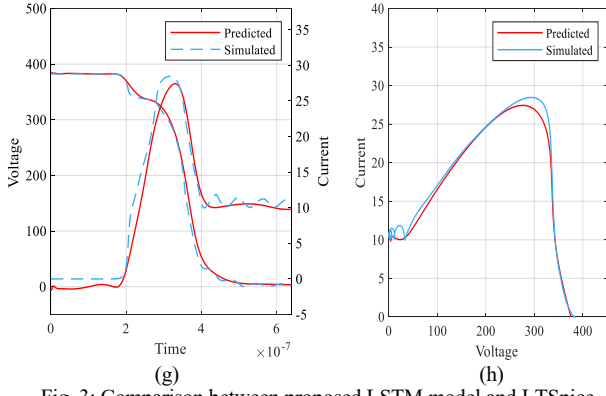


Fig. 3: Comparison between proposed LSTM model and LTSpice simulation. (a,b) set 1; (c,d) set 2; (e,f) set 3; (g,h) set 4 as per table 4.

The Fig. 3 presents the LTSpice simulation results compared to the LSTM results. The RMSE error in predicting the switching behavior is approximately 8 to 9%. The peak current and voltage obtained from the switching transitions are substituted into (1) to calculate the damage accumulation per switching cycle.

In mission profile-based reliability analysis, considering a battery charging profile typically requires several hours to charge the battery. Calculating the reliability for one charging cycle through actual switch simulation would therefore demand substantial time and storage to determine the damage accumulation for one charging cycle. However, the proposed LSTM model replaces the actual switch simulation, enabling rapid calculation of the switch's damage accumulation for one charging profile. This is one of the applications of the proposed method.

IV. EXPERIMENTAL RESULTS

A synchronous buck converter is chosen for the verification of the proposed LSTM model. The prototype is developed and shown in Fig. 4 by considering the converter specifications presented in Table IV. The complementary pulses are generated with sufficient dead time with the help of TMS320F28335 micro-controller. The converter is operated at different power level in order to validate the proposed method. The power level is varied by changing the output current and keeping constant the DC bus voltage. Since, in many applications, as per example in battery charging applications, the load current varies but the DC bus

TABLE IV
CONVERTER SPECIFICATIONS

Parameters	Value
Input voltage (V_{in})	120 V
Switching frequency (f_{sw})	100 kHz
Inductance (L)	400 μ H
Gate resistance (R_g)	20 Ω
Switch Q_1 and Q_2	GS66508B
Gate Driver	ADUM4121ARIZ

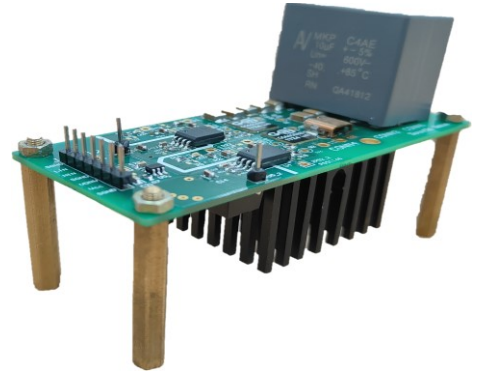


Fig. 4: GaN based converter prototype

voltage remains constant. The spurious firing is avoided by properly utilizing the miller clamp pin of the driver and minimizing the gate of the loop of the switch.

A. PCB parasitic extraction:

The proposed model is dependent on the PCB stray inductance values. To validate the proposed method, the PCB parasitic extraction is necessary of the designed prototype as shown in Fig. 4. The stray inductances are calculated with the help of Ansys Q3D. The PCB layout is designed in Altium Designer. Then the actual layout is imported to the Ansys Q3D and simulated in order to extract the stray inductances. The extracted inductances are presented in Table V. As it is visible from the Table V that the stray inductance values are in the range presented in Table II.

TABLE V: PARASITIC ELEMENTS AND THEIR VALUES

Elements	Extracted Values
L_{d1}	0.99nH
L_{s1}	0.12nH
L_{d2}	0.2nH
L_{s2}	0.98nH
L_{g1}	0.29nH
L_{g2}	0.3nH

The effect of L_{d1} , L_{s1} and L_{d2} , L_{s2} are lumped and initialized in the proposed LSTM model to generate the switching transition. These stray inductances are one of the components of the feature matrix (x_t) as described in Section II.

B. Validation of proposed model:

The model validation is conducted under two different operating conditions. First, the prototype is operated with a DC bus voltage of 50V and a power level of 100W. Second, it is operated with a DC bus voltage of 120V at a power rating of 200W. The comparison between the experimental results and the LTSpice results is presented in Fig. 5.

The Fig. 5 demonstrates the ability of the trained LSTM model to estimate the (v_{ds}) transition accurately. It is found that the model output closely matches the measured results.

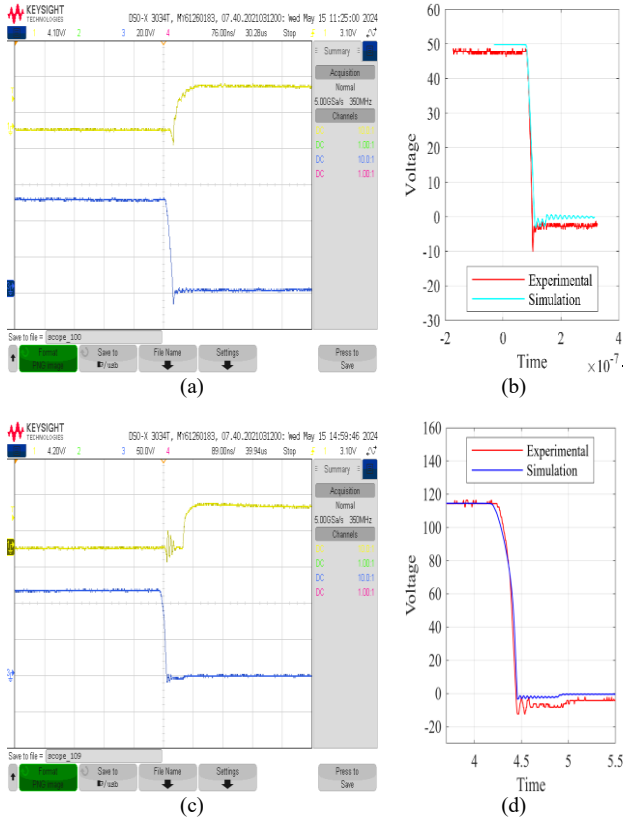


Fig. 5 (a,c): V_g and V_{ds} plot for 100 W and 200 W respectively & (b,d): Comparison of V_{ds} between experimental and simulation results for 100W and 200 W respectively.

Combining the results from Figs. 3 and 5, the LSTM model can accurately predict the switching behavior of the GaN devices in a very short time.

The validation of the switching current transition (i_{ds}) will be included in the revised version of the paper due to the unavailability of the current shunt (SDN-414-10).

V. CONCLUSIONS

In this paper, LSTM is used to enhance GaN switch transition modelling for converters. This finding highlights the LSTM's potential in accurately predicting switching waveforms and capturing device behaviour, thereby contributing to the improved performance and efficiency of power electronic systems. The effective utilization of the LSTM model presents promising opportunities to optimize power electronic systems, meeting the increasing demand for high-efficiency, high-performance devices across various applications such as renewable energy, electric vehicles, and consumer electronics. The LSTM model's capability to adapt and learn from data makes it a powerful tool for advancing technology in power electronics. However, it is important to note that in this paper, only a limited number of parasitic elements are considered, suggesting that further research is needed to include more comprehensive modelling.

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